MotionFollower: Editing Video Motion via Score-Guided Diffusion

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https://github.com/Francis-Rings/MotionFollower

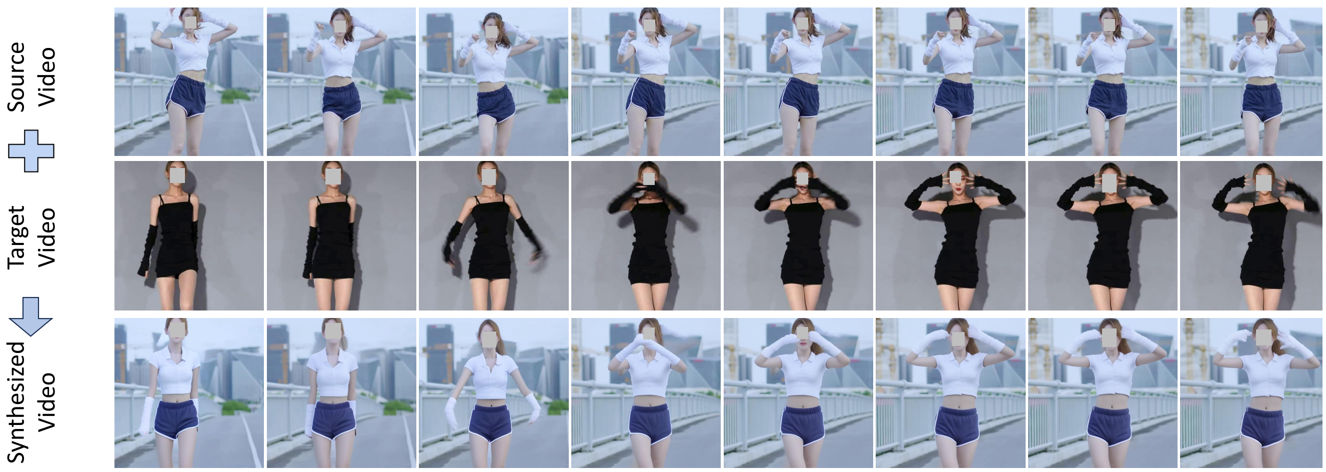

Figure 1. MotionFollower: a motion editing method for transferring motion from target video to source while keeping the source background, protagonists' appearance, and camera movement.

Abstract

Despite impressive advancements in diffusion-based video editing models in altering video attributes, there has been limited exploration into modifying motion information while preserving the original protagonist's appearance and background. In this paper, we propose MotionFollower, a score-guided diffusion model for video motion editing. To introduce conditional controls to the denoising process, we propose two signal controllers, one for poses and the other for appearances, both consist of convolution blocks without involving heavy attention calculations. Further, we design a score guidance principle based on a two-branch architecture (a reconstruction and an editing branch), significantly enhancing the modeling capability of texture details and complicated backgrounds. Concretely, we enforce several consistency regularizers during the score estimation. The resulting gradients thus inject appropriate guidance to latents, forcing the model to preserve the original background details and protagonists' appearances without interfering with the motion modification. Experiments demonstrate MotionFollower's competitive motion editing ability qualitatively and quantitatively. Compared with MotionEditor, the most advanced motion editing model, MotionFollower delivers superior motion editing performance and exclusively supports large camera movements. To the best of our knowledge, MotionFollower is the first diffusion model to explore score regularization in video editing.

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1. Introduction

Diffusion models [7, 12, 13, 25, 27, 28, 31, 35, 39, 41] have demonstrated their superiority in performing image and video generation [14, 31, 46], which promote numerous studies in video editing [3, 5, 9, 19, 20, 29, 44, 47]. While considerable progress has been made, current models predominantly focus on attribute-level editing, such as video style transferring and manipulation for background and protagonist's appearance. Despite being recognized as one of the most distinct and sophisticated features compared to images, motion information is largely overlooked in existing models. Inspired by MotionEditor [40], we aim to manipulate the motion of a video in alignment with a target video, which is regarded as a higher-level and more challenging video editing scenario—motion editing. In particular, given a target video and a source video, we wish to transfer the motion information of the target video to the protagonist in the source while preserving the original details.

As of current literature, MotionEditor [40] is the pioneer work capable of editing motions, which leverages an adapter appended to ControlNet [54] and an attention injection mechanism. However, it presents a degrading performance regarding temporal consistency and suffers from substantial computation costs stemming from the ControlNet and the attention injection. Moreover, the performance of MotionEditor deteriorates when encountering specific scenarios, such as large camera movements and complex backgrounds. Recent studies have also explored human motion transfer [16, 23, 33, 43, 52, 57] and pose-driven video generation [24, 56]. The former aims to animate a given image based on target signals, while the latter generates signal-guided videos without preserving the desired appearance. Video motion editing has a considerable discrepancy, primarily manipulating the video's motion while maintaining other details, such as camera movements, per-frame background variations, and the protagonist's appearance.

We propose a novel editing approach, MotionFollower, to tackle the above issues as summarized in Fig. 2. We design two signal controllers (Pose Controller and Reference Controller) to enhance the capabilities of controlling pose and modeling appearance. These two controllers comprise pure convolutions without costly attention calculations, considerably reducing the computations.

We further propose a novel score-guided principle that impels the model to keep the original semantic details during inference, *e.g.* background and camera movement. Instead of injecting attention maps or interacting with source key/value as in previous methods, which may cause overlapping and shadow flickering, we introduce score regularization to improve the consistency between source and edited video. The denoising process can be regarded as a continuous process [36], which can be described as the score function. This score function steers the denoising towards a specific direction. Thus, we can influence the model to denoise data according to our preferences by incorporating additional regularization and thereby ensuring the outputs retain the essential semantic details of the source video *e.g.*, background and camera movement. Specifically, with a reconstruction branch and an editing branch, we utilize segmentation masks to calculate multiple regularization terms corresponding to different regions. Guided by the score function, the latent from U-Net is then optimized by gained gradients, facilitating regional consistency between the editing and reconstruction branches. Our guidance principle does not update the weights of the trained model.

In conclusion, our contributions are as follows: (1) We propose a novel score guidance principle, enabling the diffusion model to preserve regional details of the source by introducing several regularizations into the score function, thereby increasing content consistency. To the best of our knowledge, MotionFollower is the first diffusion model to explore score regularization in video editing. (2) We introduce two signal controllers to replace the heavy ControlNet, without losing control ability and performance. (3) Experiments demonstrate the superior performance of MotionFollower, particularly on specific videos featuring large camera movements and complex backgrounds.

2. Related Work

Diffusion for Video Editing and Generation. The success of diffusion-based image generation has inspired numerous studies on diffusion-based video generation [8, 14, 15, 19, 21, 34, 37, 38, 48–50]. Most T2V models inflate pretrained T2I diffusion models by inserting additional temporal layers. AnimateDiff [10] proposes a motion module that can be directly plugged into other personalized diffusion models. Meanwhile, pose-guided video synthesis is particularly prevalent. Follow-Your-Pose [24] and ControlVideo [56] insert an adapter into the T2I diffusion model for modeling the pose sequence. However, the video generation models mainly focus on diversity instead of preservation.

Most video editing approaches can be divided into two categories, namely the mutual attention-based method and the layered neural atlas-based method. Tune-A-Video [47], Text2Video-Zero [19], FateZero [29], CCEdit [9] belong to the first class, while Text2LIVE [2], StableVideo [4], and VidEdit [6] belong to the latter. However, the above methods primarily focus on low-level attribute editing and are unable to edit complex motions. Regarding video motion editing, MotionEditor [40] stands as the most advanced diffusion model equipped for this task, achieved through applying a motion adapter and a complex attention injection, escalating its computational demands. Further, its attention injection struggles with certain videos with complex backgrounds or camera movements.

Human Motion Transfer. This task aims to animate images based on one given image. Previous GAN-based models [17, 22, 23, 32, 33] struggle with complex backgrounds and poses. Recently, some studies have applied diffusion models to this field. DisCo [43] uses masks to decouple the objects for modeling. MagicAniamte [52] leverages ControlNet and an additional U-Net for modeling poses and appearances independently. AnimateAnyone [16] combines attention features from the reference net with those of a

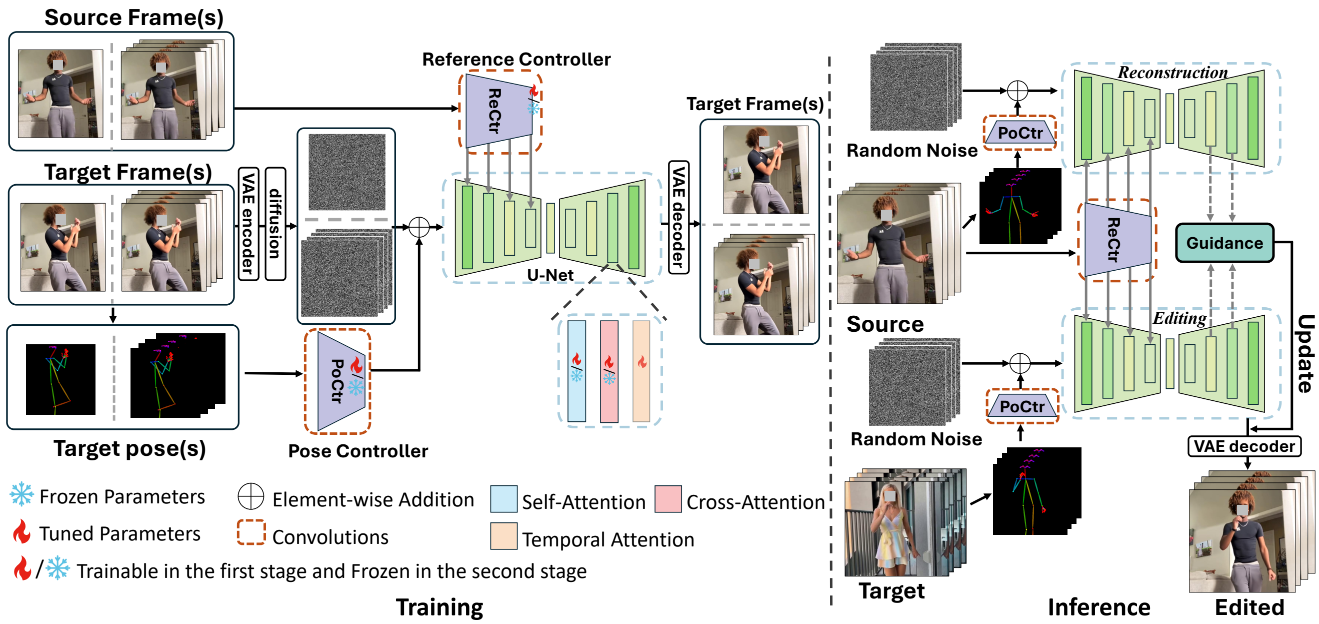

Figure 2. Architecture overview. In training, two lightweight signal controllers and U-Net are trainable. The model is first trained with the single frame (first stage), then with video clips (second stage). In inference, we build a two-branch structure, one for reconstruction and the other for editing. Score guidance is computed using features from the two branches, which is then used to update the latent.

denoising U-Net for joint self-attention. Champ [57] introduces SMPL for enhancing controlling. However, these methods are unable to handle complex motions while preserving the original camera movements and appearances. They also require the input image to have a clear initial pose, which may not be available in practice. By contrast, MotionFollower synthesizes consistent videos in complex motion while preserving original dynamic information.

3. Method

As illustrated in Fig. 2, we build MotionFollower based on a T2I diffusion model (*i.e.* LDM [31]). We improve the U-Net architecture by integrating motion modules (temporal layers) [10], inflating it to a 3D U-Net. To support conditional signals, we propose two conditional controllers: a Pose Controller (PoCtr) and a Reference Controller (ReCtr), which encode the target poses and source appearance respectively, as detailed in Sec. 3.1. Compared to ControlNet [54], our controllers are more parameter-efficient, using only a few convolution layers instead of attention layers.

During inference, MotionFollower transfers the motion from the given target video to the source video while preserving the protagonist's appearance and the background. This is achieved through a two-branch architecture, consisting of a reconstruction branch and an editing branch, both utilizing the same conditional controllers. Both branches take random noise as the initial input, and we also compute poses from the source and target videos as inputs to the PoCtr for reconstruction and editing as required. To further keep the details in the background and foreground, we propose multiple functions to force the consistency between edited and reconstructed features, encapsulated into a regularization term in the score function, as detailed in Sec. 3.3.

3.1. Conditional Controller

Pose Controller. While ControlNet [54] performs well in conditioned generation, it contains numerous costly attention operations and compromises its robustness when directly generating modified motion [40]. Therefore, we propose a lightweight module that effectively leverages pose guidance during denoising. Pose Controller (PoCtr) consists of four convolution blocks, each with two convolution layers. Pose signals are encoded into representations of the same dimension as the initial latents, which are then added to the latents before denoising. This approach is lightweight yet effective compared to conventional signal fusion strategies such as cross-attention and concatenation. We initialize the weights of PoCtr's final projection layer to zero to avoid large influences in training.

Reference Controller. To preserve the source's appearance, we introduce the Reference Controller (ReCtr) as a lightweight alternative to the time-consuming DDIM inversion [35] that can cause significant infidelity [26]. Similar to PoCtr, ReCtr includes four convolution blocks for downsampling source features to the same dimension as the latents, with each block containing two convolution layers. Further, we effectively inject the multi-scale source features into the U-Net by adding to the features of each block in its encoder. Existing comparable designs, *e.g.* MagicAnimate [52] and AnimateAnyone [16], both leverage the U-

Net as the reference controlling module, including numerous complicated attention operations. Additionally, both of them implement concatenation operations for introducing source features to denoising U-Net, increasing the channel dimension during attention calculation, thereby significantly boosting the computation consumption.

To maintain precise consistency and low-level details, we do not use textual prompt-based cross-attention. Instead, source frames are transformed into CLIP embedding using CLIPProjector [30], and subsequently integrated into U-Net by performing cross-attention with the latents [16].

3.2. Training Strategy

The training process contains two stages, *i.e.*, the image training stage and the video training stage. In the first stage, we utilize individual images for training, in which the motion modules [10] are temporarily removed to adapt to the image domain. Both the source and target images are randomly extracted from the same video clip. Two controllers and U-Net are trainable in this stage. The weights of two controllers are initialized from Gaussian and the U-Net is based on the pretrained SD1.5 [31]. The objective of this stage is to enable MotionFollower to perform image editing given a source image and a target pose.

In the second stage, we incorporate motion modules, *e.g.*, temporal layers, into the above-trained MotionFollower for adapting to videos. The temporal layers are trainable while fixing the rest of MotionFollower. Following previous practices [16, 52, 57], the weights of temporal layers are initialized by pretrained AnimateDiff [10]. It is noticeable that training temporal layers solely on video clips may deteriorate the model to perform high-quality conditioned editing. To address this issue, we conduct a hybrid training scheme, including image-based editing and video-based editing. With a probability of 40%, we sample two images as in the first stage for image editing training to maintain single-frame fidelity. Otherwise, we randomly select two distinct video clips from the same source video for training, with one acting as the source and the other as the target. This stage significantly enhances the temporal coherence of MotionFollower.

3.3. Consistency Guidance via Score Regularization

Although extra signals (target poses and source frames) help control the structure and appearance, the preservation of foreground and background details is not assured due to the lack of supervision in end-to-end training. The reversed diffusion process (*i.e.*, the denoising process) can be derived once the score of the distribution (*i.e.*, the gradient of the log probability density w.r.t. data) is known at each time step [12, 36]. The score function can guide the denoising process in a particular direction. This indicates we can add constraints by imposing regularization on the score when estimating it to force the model to denoise the data in our desired direction, thereby solving the inconsistency.

Specifically, given the intermediate feature $\boldsymbol{F}_t^r, \boldsymbol{F}_t^e$ from the reconstruction and editing branches at the t-th step, we formulate our target score function by introducing consistency guidance as follows:

$$\nabla_{\boldsymbol{z}_t^e} \log q(\boldsymbol{z}_t^e, \boldsymbol{F}_t^e, \boldsymbol{F}_t^r) = \nabla_{\boldsymbol{z}_t^e} \log q(\boldsymbol{z}_t^e) + \nabla_{\boldsymbol{z}_t^e} \log q(\boldsymbol{F}_t^e, \boldsymbol{F}_t^r \mid \boldsymbol{z}_t^e), \tag{1}$$

where $\boldsymbol{z}_t^e$ is the latent in denoising U-Net of the editing branch, and $q(\cdot)$ is the density distribution. The first term on the right-hand side is the original score for denoising, and the second term $\nabla_{\boldsymbol{z}_t^e} \log q(\boldsymbol{F}_t^e, \boldsymbol{F}_t^r \mid \boldsymbol{z}_t^e)$ is the consistency regularization, which aims to guide the denoising process in the desired direction, ensuring the generated results to have consistent appearance as the source.

To refine both foreground and background quality in edited results, we design two functions $\mathcal{S}_{fg}$ and $\mathcal{S}_{bg}$ (consisting of three sub-terms: the background overlapping term $\mathcal{S}_{over}$, body non-overlapping term $\mathcal{S}_{body}$, and complementary term $\mathcal{S}_{com}$), whose variables are $\boldsymbol{F}_t^e$ and $\boldsymbol{F}_t^r$, to approximate the score $\nabla_{\boldsymbol{z}_t^e} \log q(\boldsymbol{F}_t^e, \boldsymbol{F}_t^r \mid \boldsymbol{z}_t^e)$. We first leverage a lightweight segmentation model (see Sec. 1 and Sec. 8 of the Supp) to obtain the source foreground mask $\boldsymbol{M}^r$ and predict the edited foreground mask $\boldsymbol{M}^e$ (*i.e.*, the mask of the source protagonist aligning with the target pose). The segmentation model takes a reference image with a pose as input to predict a mask aligning with the given reference and pose. $\boldsymbol{M}^r$ is obtained by sending the source frame and source pose to the segmentation model, while $\boldsymbol{M}^e$ is predicted by sending the source frame and the target pose. Then, the score can be converted to $\frac{d(\mathcal{S}_{fg} \odot \boldsymbol{M}^e + \mathcal{S}_{bg} \odot (1-\boldsymbol{M}^e))}{d\boldsymbol{z}_t^e}$. The fg and bg in the $\mathcal{S}_{fg}$ and $\mathcal{S}_{bg}$ refer to protagonists and backgrounds in the edited results, respectively. Utilizing $\boldsymbol{M}^e$ to combine $\mathcal{S}_{fg}$ and $\mathcal{S}_{bg}$ aims at performing precise regional updates on the latents concerning foreground and background.

Fig. 3 depicts the functions above (we use RGB images to represent the feature maps for better illustration). Concretely, $\mathcal{S}_{fg}$ aims to maximize the similarity between the foreground features of the editing and reconstruction branches, enforcing the model to keep the protagonist's appearance:

$$\mathcal{S}_{fg}(\boldsymbol{F}_t^e, \boldsymbol{F}_t^r) = \frac{\alpha_{fg}}{1 + 2 \cdot \mathtt{Sim}(\mathtt{Pool}(\boldsymbol{F}_t^e \odot \boldsymbol{M}^e), \mathtt{Pool}(\boldsymbol{F}_t^r \odot \boldsymbol{M}^r))}, \tag{2}$$

where $\mathtt{Sim}(\cdot)$ and $\mathtt{Pool}(\cdot)$ refer to spatially averaged cosine similarity and mask pooling [51]. α_{fg} is a hyper-parameter. The spatial sizes of $\boldsymbol{F}_t^e$, $\boldsymbol{F}_t^r$, $\boldsymbol{M}^e$, and $\boldsymbol{M}^r$ are interpolated into the same size $H \times W$, *i.e.*, $\boldsymbol{F}^e, \boldsymbol{F}^r, \boldsymbol{M}^e, \boldsymbol{M}^r \in \mathcal{R}^{T \times N \times H \times W}$. N is the number of frames in the video clip and T is the total number of denoising steps.

$\mathcal{S}_{over}$ impels the diffusion model to reconstruct overlapping background between the editing and reconstruction features during denoising:

$$\begin{aligned} \boldsymbol{M}^{over} &= (1-\boldsymbol{M}^{e}) \odot (1-\boldsymbol{M}^{r}), \\ \mathcal{S}_{over}(\boldsymbol{F}_t^e, \boldsymbol{F}_t^r) &= \frac{1}{1+2\cdot \mathtt{Avg}(\sum_{k,i,j,\boldsymbol{M}^{over}_{t,k,i,j}=1} \mathtt{Cos}(\boldsymbol{F}^e_{t,k,i,j}, \boldsymbol{F}^r_{t,k,i,j}))}, \end{aligned} \tag{3}$$

where $\boldsymbol{M}^{over}$ refers to the overlapping background regions between the source and the edited videos. $\mathtt{Avg}(\cdot)$ indicates the average operation and $\mathtt{Cos}(\cdot)$ is the cosine similarity. Notably, directly copying overlapping pixels from the source may create overly sharp boundaries, whereas $\mathcal{S}_{over}(\cdot)$ produces more natural results.

When ReCtr sends the appearance feature extracted from the source to the editing branch, it potentially introduces source pose bias into the feature distribution. Thus, poses that align with the source rather than the target may lead to the appearance of source poses in areas that should be the background in the edited results. We use $\mathcal{S}_{body}$ to mitigate the pronounced ghosting and blurring effects due to such motion misalignment between the source and the target:

$$\begin{aligned} \boldsymbol{M}^{body} &= \boldsymbol{M}^{r} \odot (1-\boldsymbol{M}^{e}), \\ \mathcal{S}_{body}(\boldsymbol{F}_t^e, \boldsymbol{F}_t^r) &= \mathtt{Avg}(\sum_{k,i,j,\boldsymbol{M}^{body}_{t,k,i,j}=1} \mathtt{Cos}(\boldsymbol{F}^e_{t,k,i,j}, \boldsymbol{F}^r_{t,k,i,j})), \end{aligned} \tag{4}$$

where $\boldsymbol{M}^{body}$ refers to misaligned motion parts (non-overlapping parts of the protagonist) between the source video and the edited video.

Finally, $\mathcal{S}_{com}$ ensures that the content of the non-overlapping protagonist's parts closely resembles the background of the entire source video. Specifically, it advances the diffusion model to inpaint non-overlapping protagonist's parts between the source video and the edited video, leveraging background information from the source:

$$\mathcal{S}_{com}(\boldsymbol{F}_t^e, \boldsymbol{F}_t^r) = \frac{1}{1+2\cdot \mathtt{Sim}(\mathtt{Pool}(\boldsymbol{F}_t^e \odot \boldsymbol{M}^{body}), \mathtt{Pool}(\boldsymbol{F}_t^r \odot (1-\boldsymbol{M}^{r})))}. \tag{5}$$

In summary, $\mathcal{S}_{bg}$ combines three sub-terms as follows:

$$\mathcal{S}_{bg} = \alpha_{over}\mathcal{S}_{over} + \alpha_{body}\mathcal{S}_{body} + \alpha_{com}\mathcal{S}_{com}, \tag{6}$$

where α_{over}, α_{body}, and α_{com} balance terms. The backbone normalizes features before score regulation for computing the cosine similarity. The consistency guidance $\nabla_{\boldsymbol{z}_t^e} \log q(\boldsymbol{F}_t^e, \boldsymbol{F}_t^r \mid \boldsymbol{z}_t^e)$ finally combines $\mathcal{S}_{fg}$ and $\mathcal{S}_{bg}$:

$$\nabla_{\boldsymbol{z}_t^e} \log q(\boldsymbol{F}_t^e, \boldsymbol{F}_t^r \mid \boldsymbol{z}_t^e) := \frac{d\mathcal{S}_{fg}}{d\boldsymbol{z}_t^e} \odot \boldsymbol{M}^e + \frac{d\mathcal{S}_{bg}}{d\boldsymbol{z}_t^e} \odot (1-\boldsymbol{M}^e). \tag{7}$$

We exploit our regularization in inference to optimize the latents rather than the U-Net. More details are in Sec. 1, Sec. 4, and Sec. 5 of the Supp.

4. Experiments

4.1. Implementation Details

We collect 3K videos (60-90 seconds) from the internet to train our model. We utilize DWPose [53] and our proposed

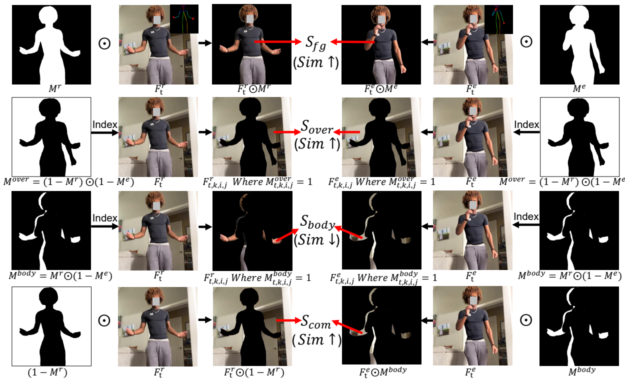

Figure 3. The illustration of our proposed functions.

lightweight person segmentation method to extract poses and masks. The details of our person segmentation method are depicted in Sec. 1 of the Supp. We evaluate our model on 100 unseen in-the-wild cases selected from YouTube and the TikTok dataset [18]. The training process is conducted on 4 NVIDIA A100 GPUs. The first and second training stages are both trained for 100,000 steps with a batch size of 64 and 4 respectively. The learning rates of the two stages are set to $1e-5$. In inference, we first align the target pose sequence with the source by introducing the pose alignment algorithm [40]. MotionFollower requires 50 seconds to perform motion editing of 24 frames on a single A100.

$\boldsymbol{F}_t$ is extracted from the second and third decoder layers as they have more semantic information and low-level details, suitable for producing guidance signals. In contrast, the first-layer features are too high-level, and the fourth-layer features lack feature correspondence, making accurate image reconstruction difficult. In experiments, we find that score guidance in later diffusion steps (higher timesteps) harms texture quality in the edited videos, so we apply our guidance only during the first n=30 steps of 50 total denoising steps. The score becomes spatially decomposable around the 30th denoising step.

4.2. Comparison with State-of-the-art Methods

Competitors. The competitors include MRAA [33], Tune-A-Video [47], Disco [43], MagicAnimate [52], AnimateAnyone [16], Champ [57], and MotionEditor [40]. Particularly, Tune-A-Video is equipped with ControlNet [54] to enable controllable editing. Human animation models (MRAA, Disco, MagicAnimate, AnimateAnyone, Champ) have been adapted to support video motion editing by sending all source frames to their reference networks instead of the original single image.

Qualitative results. We conduct a qualitative comparison between our MotionFollower and several competitors in Fig. 4. By analyzing the results, we can gain the following observations: (1) All competitors can manipulate the motion of the given video to some extent. However, their edited results exhibit significant ghosting issues, where

Figure 4. Qualitative comparison between our MotionFollower and other state-of-the-art models.

frames of the protagonist's heads and legs overlap, creating strong artifacts. (2) Tune-A-Video [47] encounters severe overlapping effects and fails to preserve the protagonist's appearance. The plausible reason is that they do not employ the interaction of the source intermediate features. (3) In terms of the human motion transfer methods (MagicAnimate [52], AnimateAnyone [16], Champ [57]), they fail to perform accurate motion manipulation and reconstruct the protagonist's clothes with high fidelity. The results indicate that existing human motion transfer models encounter challenges when performing motion editing on videos containing complex information. (4) The most advanced video motion editing model, MotionEditor, struggles with body distortion. (5) Our MotionFollower excels at precisely modifying motion in videos while maintaining the integrity of the source background and appearance, showcasing its significant superiority over previous methods.

Quantitative results. We partitioned a test video into two segments: the first segment serves as the input, while the second segment serves as the target. The reason is that we cannot obtain the ground truth of cross-id edited videos, as it is challenging to find two videos with different identities performing the same motion. We conduct the validation on 100 collected in-the-wild videos, comprising complex backgrounds and protagonists' appearances. Table 1 depicts the quantitative comparison between our method

Model	L1 ↓	PSNR* [43] ↑	SSIM [45] ↑	LPIPS [55] ↓	FID [11] ↓	FID-VID [1] ↓	FVD [42] ↓	Mem ↓
MRAA [33]	7.83E-4	8.03	0.35	0.67	120.51	100.12	783.72	**6.4G**
Tune-A-Video [47]	3.88E-4	12.01	0.42	0.51	66.37	84.58	613.44	31.7G
Disco [43]	2.92E-4	9.24	0.38	0.47	92.97	83.69	635.08	18.3G
MagicAnimate [52]	1.09E-4	16.22	0.62	0.35	33.04	26.59	477.65	21.8G
AnimateAnyone [16]	9.45E-5	16.18	0.65	0.32	35.81	28.31	515.57	16.1G
Champ [57]	9.94E-5	16.12	0.58	0.36	36.55	25.89	452.65	17.4G
MotionEditor [40]	9.13E-5	17.34	0.68	0.34	31.98	20.57	395.43	42.6G
MotionFollower	**6.31E-5**	**20.85**	**0.75**	**0.22**	**26.30**	**12.42**	**276.04**	9.8G

Table 1. Quantitative comparison on 100 wild cases. Mem refers to GPU memory when manipulating 8 frames.

and other competitors. We find that MotionFollower outperforms all competitors in terms of single-frame quality and video fidelity at a cost-efficient rate. Our model outperforms the best competitor MotionEditor by achieving 8.15 FID-VID and 119.39 FVD with an approximately 80% reduction in GPU consumption, demonstrating the superiority of our model among other competitors. With a single NVIDIA A100, MotionEditor can only edit 16 frames at a time, while our MotionFollower can edit 56 frames.

4.3. Discussion

Multiple Motion Types. Our MotionFollower aims to address human video motion editing. We conducted an experiment that considered a wide range of motion, such as running, playing Tai Chi, and playing basketball, as shown in Sec. 9.1 of the Supp. We can see that our MotionFollower can handle a wide range of motion types.

Long Video Synthesis. We conduct a qualitative comparison between our proposed MotionFollower and the strongest competitor MotionEditor in Sec. 9.2 of the Supp. The source video comprises 600 frames containing intricate clothes, complex initial poses, and dynamic camera movements. The results depict that the edited frames of MotionEditor encounter numerous blurry noises and body distortion. A possible reason is that camera movements introduce dynamism to objects and backgrounds, making it challenging for conventional diffusion models to capture semantic information accurately. By contrast, due to our consistency guidance, our model can perform precise motion editing on long videos with dynamic camera movements while preserving the original information.

Camera Movement. Our consistency guidance can force the diffusion model to focus on the semantic details of the source, thereby promoting the model to preserve the original background information, including camera movements. Results are shown in Sec. 9.3 of the Supp. Large camera movements can deteriorate the editing capability of all competitors, as they struggle to maintain the semantic details of the dynamic background, thereby exacerbating the occurrence of blurring noise. In contrast, our MotionFollower can effectively perform video motion editing while preserving complex camera movements and background details.

Human Motion Transfer. Video motion editing conducts

Model	L1 ↓	PSNR* ↑	SSIM ↑	LPIPS ↓	FVD ↓
MRAA [33]	3.21E-4	29.39	0.672	0.296	284.82
Disco [43]	3.78E-4	29.03	0.668	0.292	292.80
MagicAnimate [52]	3.13E-4	29.16	0.714	0.239	179.07
AnimateAnyone [16]	-	29.56	0.718	0.285	171.9
Champ [57]	3.02E-4	**29.84**	0.773	0.235	170.20
MotionFollower	**2.89E-4**	29.25	**0.793**	**0.230**	**159.88**

Table 2. Comparison on the Human Motion Transfer task.

Model	Optical Flow L2 ↓	M-C	T-C	A-C
MagicAnimate	7.21	92.5%	90.4%	94.6%
AnimateAnyone	7.65	95.7%	93.5%	96.8%
Champ	7.43	94.2%	93.7%	95.3%
MotionEditor	5.87	91.3%	88.6%	92.4%
MotionFollower	**3.21**		-	

Table 3. Consistency validation results. M/T/A-C refer to Motion/Temporal/Appearance Consistency. The last three columns are the user preference ratio of our model compared with other methods. Higher indicates the users prefer more to our model.

motion transfer directly on the source video over the temporal dimension, which involves multiple initial poses, camera movements, and dynamic backgrounds. This task is typically more challenging than human motion transfer, which only focuses on animating a single image with a simple initial pose and a static background while neglecting dynamic information. To demonstrate that MotionFollower also excels over other methods in this less challenging task, we convert the workflow of motion editing into the workflow of motion transfer for a comparative analysis of motion manipulation capabilities. We follow the evaluation as DisCo [43], training and testing our model on TikTok [18] for fair comparisons. The quantitative comparisons are presented in Table 2. Our method achieves the best results in most metrics, showing that our model can effectively handle image animation due to the power of motion manipulation.

We also show a qualitative comparison between our MotionFollower and other animation models regarding motion transfer, as presented in Sec. 9.4 of the Supp. Our model can perform gorgeous image animation, while the results of other competitors suffer from body distortion and a dramatic degree of ghosting from overlapping hands.

Consistency Validation. There is currently no widely ac-

Model	PSNR* ↑	SSIM ↑	LPIPS ↓	FVD ↓	Mem ↓
w/o ReCtr	9.03	0.42	0.45	545.39	8.3G
(*w/o* ReCtr)+*RNet*	19.82	0.72	0.25	288.57	17.8G
(*w/o* PoCtr)+*CNet*	18.13	0.65	0.28	381.52	15.2G
w/o S_{fg}	18.24	0.66	0.31	372.34	**7.1G**
w/o S_{over}	19.47	0.69	0.27	312.24	**7.1G**
w/o S_{body}	20.31	0.71	0.25	285.17	**7.1G**
w/o S_{com}	20.52	0.72	0.24	280.28	**7.1G**
w/o score guidance	16.50	0.61	0.36	437.69	**7.1G**
Ours	**20.85**	**0.75**	**0.22**	**276.04**	9.8G

Table 4. Ablations on key components. *+RNet* and *+CNet* refers to inserting the Reference Net (RNet) from MagicAnimate and ControlNet (CNet) into our model, respectively.

cepted metric for evaluating the consistency of videos. Inspired by [4], we first partition a test video into two segments: the first acts as the input, while the second acts as the target. We then calculate the average L2 distance of the dense optical flow [4] between the edited videos and target videos. we also conduct a user study considering consistency on 100 videos. For each case, participants are first presented with the source and target with different identities and motions. Then, we present two videos in a random order (one edited by MotionFollower and the other by a competitor). The results are shown in Table 3. We can see that our model has the best capability to maintain consistency.

4.4. Ablation Study

ReCtr and PoCtr. In Table 4 and Fig. 5. We replace our proposed ReCtr and PoCtr with Reference Net (RNet) in MagicAnimate [52] and ControlNet (CNet) [54], respectively. We can observe that *w/o* ReCtr has a significantly negative impact on the editing performance, as the model solely relies on cross-attention with source embedding. (*w/o* ReCtr)+*RNet* leads to slightly inferior performance compared to our model. Moreover, the UNet-based RNet consumes roughly twice the GPU memory as our model. In the third row of Fig. 5, the protagonist's clothes are considerably blurry compared to our model, indicating that our ReCtr can retain more appearance details due to pure convolution operations. Similarly, (*w/o* PoCtr)+*CNet* can also exacerbate the strain on GPU memory consumption, as ControlNet consists of multiple costly attention operations. We can also observe that replacing our PoCtr with ControlNet degrades video quality. The plausible reason is that attention-based conditional models lack inductive bias, requiring large amounts of data to learn prior information. Our collected training dataset is insufficient to meet the data demands of attention-based conditional models. By contrast, our controllers are easier to possess an inductive bias due to their pure convolution layers, enabling the model to achieve optimal performance even with limited data.

Score Guidance. In Table 5, *w/o* S_{fg} and *w/o* S_{over} both

Figure 5. Ablation study on core components of MotionFollower.

Model	SSIM↑	FID-VID↓	FVD↓
w/o Masks	0.53	24.87	442.34
w/o M^{over}	0.70	18.13	327.21
w/o M^{body}	0.72	16.28	308.10
w/o M^e&M^r in Eq.2	0.62	23.71	394.56
Ours	**0.75**	**12.42**	**276.04**

Table 5. Ablations on different masks. *w/o* Masks removes all the masks used in the score guidance.

significantly affect the performance, indicating that S_{fg} and S_{over} force the diffusion model to preserve original contents. By contrast, *w/o* S_{body} and *w/o* S_{com} result in slight deterioration in quality, showing that S_{body} and S_{com} handle motion misalignment and appearance overlapping issues, thus improving performance to some extent. In Fig. 5 (5th row), *w/o* score guidance suffers from a blurry background and disappearance of semantic details, particularly in the last two frames due to camera movement, showing that our guidance can drastically retain the content consistency. We further experiment on different masks used in the score guidance, as shown in Table 5. The results show that our model performs better as more masks are used.

5. Conclusion

In this paper, we propose MotionFollower for addressing the video motion editing challenge, which remains relatively unexplored and is considered high-level video editing compared to conventional video attribute editing. To enhance the signal-controlling capability, we propose two modules ReCtr and PoCtr (one for source and the other for poses), which purely consist of convolution operations. We further propose a score guidance principle, empowering the diffusion model to accurately reconstruct both the backgrounds and the details of the protagonist's appearance. Experiments show that our model has the superiority of motion manipulation while preserving original details.

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